UG (NEP) 6th SEMESTER
APPLIED COMPUTING (ARTIFICIAL INTELLIGENCE)
(ACP622N)

Session: June 2025 Date of Exam: 27-06-2025
Time Allowed: 2:00 Hours Maximum Marks: 75 Minimum Marks: 27
No. of Questions: 09

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

SECTION A: Short Answer Type Questions (8 × 1.5 = 12 Marks)

Answer all questions in a maximum of 20 words each.

1. What is Artificial Intelligence?
2. What is meant by Turing test?
3. Define Rational Agent.
4. What is the full form of PEAS in AI context?
5. Define state space in the context of Artificial Intelligence.
6. What is informed search?
7. State the Modus Ponens rule.
8. What is conditional probability?

SECTION B: Medium Answer Type Questions (4 × 7.5 = 30 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. What is an agent? Explain the basic kinds of agent program.

OR

Discuss the historical development of AI. Mention key milestones.

3. Compare informed and uninformed search strategies in AI with examples.

OR

What is heuristic search? Explain the Best First Search algorithm.

4. Define Hill Climbing search. Differentiate simple Hill Climbing and Steepest-Ascent Hill Climbing.

OR

What is First-Order Logic? Define the syntactic elements of First-Order Logic.

5. State Bayes’ Rule and explain how it is used to update beliefs.

OR

Explain prior probability and conditional probability with examples.

SECTION C: Long Answer Type Questions (2 × 16.5 = 33 Marks)

Answer only two questions in a maximum of 400 words each.

6. Explain in detail the structure of different intelligent agents.
7. What is A* search? Explain various stages of A* search with an example.
8. (a) Show that the following is a valid argument: “If it is humid then it will rain, and since it is humid today it will rain.”
   (b) Compute the truth value of α: (A ∨ B) ∧ (~B → A) using the truth-table approach.
9. Explain Dempster–Shafer theorem and its significance in handling uncertainty.